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 Chapter 6 Conclusion

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CHAPTER 6

CONCLUSION

6.1 Introduction

This chapter concludes the report on the Al-Muneer Online Portal project. The project progressed through the PSM1 planning and design phase and into the PSM2 implementation phase, resulting in a functional web-based platform for Al-Muneer Hall. Chapter 5 presented the implementation results, key system interfaces, and testing activities. This chapter summarises the overall achievement of the project objectives, reflects on the outcomes, and identifies possible directions for future improvement and maintenance.

6.2 Achievement of Project Objectives

The original project objectives have been addressed throughout the project lifecycle. Requirements elicitation and analysis were completed in PSM1 through stakeholder engagement with Mr. Ahmed Almunajid, producing a comprehensive Software Requirements Specification (SRS). System architecture and design were documented in the Software Design Description (SDD), including the MVC architecture, database schema, and UI/UX wireframes.

During PSM2, the software development and implementation objective was realised by building the portal with Spring Boot, PostgreSQL, Thymeleaf, and JavaScript. All core modules were implemented: visitor information browsing, media gallery with label filters, interactive availability calendar, dynamic pricing display, booking inquiry submission with 9-digit reference codes, payment proof upload, feedback submission, JWT-secured administrator authentication, content management, inquiry and payment management, WhatsApp notification templates, calendar management, dashboard analytics, reports with charts, and Gemini AI advisors.

Comprehensive system testing and validation were performed through black box, white box, and user testing, as detailed in Chapter 5 and the Software Test

Documentation (STD). The system was found to operate reliably and to meet the functional and non-functional requirements. Deployment to a Cloud VPS was completed, making the portal accessible via standard web browsers. Finally, technical documentation, user guidance, and maintenance considerations were prepared to support future upkeep.

6.3 Future Improvements

Although the current implementation satisfies the defined scope, several enhancements could be pursued in future iterations:

- **Direct Online Payments:** Integrating a formal payment gateway would remove reliance on manual transfer screenshots and provide immediate booking confirmation.
- **WhatsApp Business API:** Replacing client-side deep-links with the official WhatsApp Business API would enable automated, two-way messaging and richer notification templates.
- **Multi-Language Support:** Adding full Arabic and English localisation would improve accessibility for a broader audience.
- **Mobile Application:** A companion mobile app could offer push notifications and a native experience for frequent users.
- **Advanced Analytics:** Extending the analytics module with conversion funnels, cohort analysis, and automated periodic reports would support data-driven decision making.
- **External Calendar Integration:** Synchronising availability with Google Calendar or similar services would reduce manual calendar maintenance.
- **Automated Backups:** Scheduled database and file backups would strengthen operational reliability and disaster recovery.

These improvements would build upon the solid foundation established by the Al-Muneer Online Portal and further modernise the hall's digital operations.